OFFICIAL SYSTEM DOCUMENTATION

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ROADWATCH

Civic Traffic Enforcement & Cryptographic Evidence Verification Ecosystem

A comprehensive technical and operational blueprint detailing multi-role permissions, high-trust video capture pipelines, real-time cloud synchronization, and human-in-the-loop police adjudication workflows.

100%	< 500ms	SHA-256
HUMAN-IN-THE-LOOP VERIFICATION	REALTIME WEBSOCKET SYNC	CRYPTOGRAPHIC PRIVACY HASHING

Target Platform: Citizen Mobile App (Expo) • Officer Command Web Portal (ReactJS)

Backend Services: Supabase Cloud • PostgreSQL • Evidence Vault S3

Target Jurisdiction: Municipal Traffic Police Departments & Citizen Observers

Date: October 2026

Status: Approved Production Architecture

Document Structure & Executive Summary

Table of Contents

SECTION	TITLE / DOMAIN	FOCUS AREA
Section 1	Executive Summary & The Civic Challenge	Problem statement & mission
Section 2	Ecosystem User Roles & Personas	Citizen, Officer, Violator
Section 3	Role-Based Access Control (RBAC) Matrix	Security permissions & visibility
Section 4	End-to-End Operational Workflow (5 Stages)	Capture, AI, Cloud, Audit, Reward
Section 5	Subsystem Technical Architecture	Mobile, Dashboard & Supabase Core
Section 6	Database Schema & Evidence Vault Specification	Postgres tables & RLS policies
Section 7	Privacy, Integrity & Anti-Fraud Engine	Trust score, CNIC hash, GPS stamp
Section 8	Dispute & Appeal Workflow Protocol	Contestation with PKR 100 bond

1. Executive Summary & Problem Statement

Modern metropolitan roads suffer from an alarming rate of traffic violations—including signal jumping, illegal wrong-way navigation, commercial lane blockage, and hazardous driving—that overwhelm municipal traffic police departments. Traditional fixed-camera CCTV infrastructures suffer from high capital expenditure, widespread blind spots, and vulnerability to occlusion.

RoadWatch solves this dilemma by turning verified citizens into active observers. Instead of relying on unreliable, unverified social media complaints or fallible autonomous AI fines, RoadWatch establishes a **high-integrity, human-in-the-loop verification pipeline**. Citizens safely capture 10-second violation clips; the client extracts multi-angle keyframes and stamps immutable GPS/orientation telemetry; Vision AI models assist with vehicle and license plate recognition; and certified Traffic Enforcement Officers review the evidence on a zero-latency web portal before any official legal citation is sanctioned.

> **Core Architectural Guarantee: Human-in-the-Loop Adjudication**
> In compliance with constitutional due process and evidence admissibility standards, **no automated fine is ever generated solely by Artificial Intelligence**. The Vision AI serves strictly as an assistive parser; every citation requires the explicit review and digital signature of an authorized Traffic Enforcement Officer.

Ecosystem User Roles & Personas

The RoadWatch ecosystem operates on three primary operational roles and one autonomous background engine. All verification, dispute adjudication, and audit logging are unified within the single Officer Command Web Dashboard.

📱 1. Citizen Reporter

`Role: reporter`

Registered everyday pedestrians, commuters, and vehicle passengers utilizing the Expo mobile app.

- Registers with Phone OTP & 13-digit Pakistani CNIC (SHA-256 hashed).
- Launches 10-second Camera HUD with live GPS and sensor stamping.
- Reviews and corrects AI-extracted license plates and claims.
- Monitors personal submissions in the sealed Evidence Vault.
- Earns civic bounty payouts (PKR) into the digital reward wallet.
- Posts PKR 100 refundable appeal bonds on contested rejections.
- Triggers emergency SOS quick-dial (Rescue 1122 / Police 15).

👮 2. Traffic Police Officer

`Role: officer`

Certified municipal enforcement personnel operating the unified Command Audit Web Portal.

- Receives real-time incoming violations via zero-latency WebSockets.
- Streams in-browser MP4 videos directly from the secure Evidence Vault.
- Examines 3 high-resolution extracted keyframes for plate legibility.
- Cross-checks satellite GPS coordinates, timestamp, and reporter notes.
- Executes 1-click `Accept` (issues fine, credits wallet, boosts score).
- Executes 1-click `Reject` with reason tag (penalizes fake reports).
- Adjudicates Citizen Dispute Appeals under the Disputed Cases room.
- Audits closed citations and exports official CSV logs for courts.
- Monitors Citizen Reporter Registry & trust reputation scores.
- Utilizes rapid keyboard hotkeys (`A` , `R` , `Space` , `J` , `K`).

🚗 3. Vehicle Offender

`Role: offender`

The registered vehicle owner or driver committing infractions on public thoroughfares.

- Subject of legal traffic citations resulting from accepted reports.
- Protected by constitutional human-in-the-loop audit safeguards.
- Receives official citations containing authenticated video & keyframe proof.
- Retains standard municipal statutory rights of appeal in traffic court.

Role-Based Access Control (RBAC) Matrix

OPERATIONAL CAPABILITY	CITIZEN (`REPORTER`)	TRAFFIC POLICE OFFICER (`OFFICER`)
Record & Submit 10s Evidence	`ALLOW`	`DENY`
Access Personal Evidence Vault	`OWN RECORDS ONLY`	`ALL CASES (PENDING & HISTORICAL)`
Earn & Withdraw Civic Bounty (PKR)	`ALLOW`	`DENY (AUDIT ACCESS ONLY)`
Realtime Pending Queue Ingestion	`DENY`	`ALLOW (LIVE WEBSOCKET STREAM)`
Accept / Reject Legal Citation	`DENY`	`ALLOW (ONE-CLICK HOTKEYS: A / R)`
File Contested Dispute Appeal	`ALLOW (PKR 100 REFUNDABLE BOND)`	`DENY`
Adjudicate Citizen Dispute Appeals	`DENY`	`ALLOW (DISPUTED CASES ROOM)`
Export Audit Records to CSV	`DENY`	`ALLOW (CASE LIST WORKSPACE)`
Monitor Reporter Registry & Trust Scores	`OWN TRUST SCORE ONLY`	`ALLOW (REGISTRY WORKSPACE)`

End-to-End Operational Workflow

The lifecycle of a traffic violation report traverses five distinct stages, transitioning seamlessly from native mobile hardware capture to cloud analysis, officer adjudication, and financial reconciliation.

1

Stage 1: Citizen Mobile Capture & Hardware Stamping

The citizen launches the native Camera HUD. The application enforces a strict **10-second recording limiter** (with a 50MB ceiling) to preserve focused evidence. Concurrently, the mobile device samples high-precision GPS coordinates, capture timestamp, and gyro orientation angles. Immediately upon recording termination, the app extracts **3 multi-angle keyframes** (`frame_1.jpg` , `frame_2.jpg` , `frame_3.jpg`) via video thumbnailing algorithms.

`Hardware HUD` `GPS Stamping` `3x Keyframes`

2

Stage 2: Vision AI Inference & Pre-Screening

Extracted keyframes are processed through Vision AI inference models. The model identifies the vehicle category (Car, Motorcycle, Commercial Bus, Rickshaw, Heavy Truck), classifies violation categories (Red Light, Wrong-Way, Illegal Parking, Triple Riding, Helmet Violation), and performs Optical Character Recognition (OCR) on vehicle registration plates. The citizen reviews the AI prediction and attaches contextual notes.

`Vehicle Classifier` `Plate OCR` `Confidence Score`

3

Stage 3: Secure Cloud Ingestion & Evidence Vault Storage

A unique RFC4122 v4 UUID is generated as the `client_capture_id` . The MP4 video file and JPEG keyframes are transmitted over encrypted TLS to Supabase Storage's private `evidence_vault` bucket. An atomic transaction inserts metadata into `public.reports` with initial status `pending_manual_review` . Supabase's Realtime engine broadcasts the new insertion event across WebSocket channels.

`UUID v4` `evidence_vault (S3)` `WebSocket Broadcast`

4

Stage 4: Officer Command Verification & Adjudication

The report instantly populates the Officer Command Dashboard's Live Pending Queue without page refresh. Cases are prioritized by severity (CRITICAL for Red Light/Wrong Way, HIGH for Reckless Driving). The officer opens the integrated Evidence Stage, plays the full MP4 video, toggles between high-res keyframes, and verifies the plate and GPS location. The officer adjudicates via hotkeys:

- **Accept Citation (`A`):** Marks report `approved` , schedules official e-challan, credits citizen bounty (PKR 250), increments citizen Trust Score (+2).
- **Reject Citation (`R`):** Dismisses report with reason tag. If deemed fraudulent, penalizes citizen Trust Score (-15).

`Hotkeys: A / R` `In-Browser Player` `Trust Score Logic`

5

Stage 5: Citizen Dispute Appeal & Re-Audit Protocol

If an honest citizen's submission is rejected, they are notified with the officer's reason. The citizen can initiate a formal **Dispute Appeal** by depositing a **PKR 100 refundable bond** and providing written contestation. The report transitions to `pending_manual_review` under the dashboard's "Disputed Cases" tab for officer re-audit. If the rejection is overturned, the PKR 100 deposit is refunded, the bounty is awarded, and the trust penalty is reversed.

`PKR 100 Deposit` `Officer Appeal Hearing` `Bond Refund`

🏛️ Subsystem Technical Architecture

ROADWATCH DISTRIBUTED ARCHITECTURE

📱 CITIZEN MOBILE APP

- **Framework:** React Native 0.86, Expo 57, React 19.
- **Key Modules:** Camera HUD, Orientation Sensor, Location Provider, Video Thumbnails.
- **Security:** Crypto SHA-256 CNIC Hashing, AsyncStorage session caching.
- **Features:** Sealed Vault, Civic Reward Wallet, Safety SOS 1122/15.

⚡ SUPABASE CLOUD CORE

- **Database:** PostgreSQL with Row-Level Security (RLS).
- **Storage:** `evidence_vault` private bucket (MP4 & JPEG).
- **Sync:** Realtime WebSocket engine (`postgres_changes`).
- **Logic:** Pl/pgSQL triggers for trust scoring and wallet balances.

🖥️ OFFICER WEB PORTAL

- **Framework:** Pure ReactJS (100% JSX/JS), Vite 6, Tailwind CSS.
- **Architecture:** Zero TypeScript/RN bridges for maximum desktop FPS.
- **Features:** Live Queue, Evidence Stage, Appeals Room, CSV Exporter.
- **Productivity:** Full keyboard shortcuts (`A`, `R`, `Space`, `J`, `K`).

Database Schema & Data Models

TABLE NAME	PRIMARY FIELDS	PURPOSE & RLS POLICY
`public.base_users`	`id (uuid)`, `phone`, `cnic_last4`, `cnic_hash`, `role`	Identity records. Citizens can read own profile; Officers have read-only audit access.
`public.reporter_profiles`	`user_id`, `trust_score`, `total_reports`, `status`	Citizen reputation tracking. Trust score (0–100) automatically updated via DB triggers.
`public.reports`	`id`, `client_capture_id`, `plate_number`, `violation_type`, `status`, `gps_lat`, `gps_lng`	Primary violations table. Citizens can insert and view own reports; Officers have full update authority.
`public.evidence`	`id`, `report_id`, `file_path`, `file_size`, `media_type`	Metadata referencing MP4 video files and JPEG keyframes stored in `evidence_vault`.
`public.reward_wallets`	`user_id`, `balance_pkr`, `pending_clearance`	Citizen bounty balances. Modified exclusively via authenticated triggers upon citation approval.
`public.fines`	`id`, `report_id`, `amount_pkr`, `citation_code`	Official traffic citations issued to offending vehicles post-officer verification.

Privacy, Integrity & Anti-Fraud Engine

1. Privacy-Preserving Citizen Identity (NADRA CNIC Digest)

To safeguard citizen reporters against retaliation, RoadWatch enforces strict privacy standards. The mobile application never stores 13-digit National Identity Card (CNIC) numbers in plaintext on device storage or databases.

- The raw CNIC is converted client-side into an HMAC-salted **SHA-256 cryptographic digest** before transmission.
- Only the **last four digits** (e.g. `••••••••••••-7412`) are preserved for citizen account reference.
- This ensures irreversible identity hashing, completely preventing identity theft or unauthorized police lookup of reporting citizens.

2. Dynamic Trust Score & Anti-Spam Game Theory

To counteract the classic vulnerability of crowd-sourced platforms—false reports, prank submissions, and grudge filings—RoadWatch deploys an automated **Dynamic Trust Scoring Engine**:

EVENT CONDITION	SCORE ADJUSTMENT	SYSTEM CONSEQUENCE
New Citizen Registration	`Baseline: 50 pts`	Standard review priority; standard submission limits.
Officer Accepts Valid Violation	`+2 pts`	Bounty credited (PKR 250); reputation elevated towards High-Trust tier.
Officer Rejects Inconclusive/Blurry	`0 pts (Neutral)`	No penalty applied; educational feedback provided to reporter.
Officer Rejects Malicious / Fake Claim	`-15 pts`	Substantial penalty; account flagged on officer registry for suspension.
Trust Score Drops Below 20 pts	`AUTOMATIC SUSPENSION`	Submission privileges revoked; requires manual appeal to reinstate.
Trust Score Reaches 85+ pts	`FAST-TRACK TIER`	Expedited review queue placement; eligible for civic reward multipliers.

3. Hardware-Bound Evidence Integrity

- **UUID v4 Evidence Keys:** Submissions are tagged with cryptographically generated RFC4122 v4 UUIDs, preventing collision or spoofing.
- **Location Fencing:** GPS satellite coordinates and compass headings are sampled exclusively during the active 10-second capture window and permanently stamped to the evidence packet.
- **Immutable Storage Vault:** Storage objects in `evidence_vault` are write-once and can only be deleted or altered by database administrator roles.

Summary of Document Purpose:
This technical document establishes the baseline architecture, security controls, and operational guidelines for the RoadWatch traffic enforcement platform. For system deployment, execute the migration scripts located in `server/supabase/migrations/` and consult the project repository.